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Holding device, cargo handling apparatus, and holding method

Abstract

According to one embodiment, a holding device includes a first holder configured to hold an upper surface of an article, and a second holder configured to hold a side surface of the article. The holding device performs at least a first operation of the first and second holders holding the article, and a second operation of only the first holder holding the article.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5044143	12/1990	Ako et al.	N/A	N/A
5564893	12/1995	Tacchi	414/797	B65G 47/902
6652014	12/2002	Schmalz	414/737	B25J 15/0616
9498887	12/2015	Zevenbergen	N/A	B25J 15/0052
9758300	12/2016	Nakamura	N/A	B65G 47/912
10226865	12/2018	Nakayama	N/A	B25J 9/1669
2013/0096713	12/2012	Takizawa et al.	N/A	N/A
2014/0179501	12/2013	Akama et al.	N/A	N/A
2016/0137435	12/2015	Tanaka et al.	N/A	N/A
2017/0106534	12/2016	Nakamoto et al.	N/A	N/A
2017/0107058	12/2016	Nakamoto et al.	N/A	N/A
2018/0207808	12/2017	Tanaka	N/A	N/A
2018/0229987	12/2017	Kaup	N/A	N/A
2021/0069902	12/2020	Nakamoto et al.	N/A	N/A
2021/0101760	12/2020	Yabe	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111703687	12/2019	CN	N/A
3 623 324	12/2019	EP	N/A
62-141000	12/1986	JP	N/A
02-062303	12/1989	JP	N/A
03-085205	12/1990	JP	N/A
04-217515	12/1991	JP	N/A
04-371500	12/1991	JP	N/A
05-004722	12/1992	JP	N/A
05-010398	12/1992	JP	N/A
07-010278	12/1994	JP	N/A
07-041175	12/1994	JP	N/A
08-175799	12/1995	JP	N/A
08-239199	12/1995	JP	N/A
08-268142	12/1995	JP	N/A
9-141588	12/1996	JP	N/A
09-309697	12/1996	JP	N/A

10-087074	12/1997	JP	N/A
2013-086914	12/2012	JP	N/A
2014-024616	12/2013	JP	N/A
2014-124798	12/2013	JP	N/A
2016-055995	12/2015	JP	N/A
2016-094280	12/2015	JP	N/A
2017-052052	12/2016	JP	N/A
2017-149565	12/2016	JP	N/A
2018-118337	12/2017	JP	N/A
2018-521931	12/2017	JP	N/A
2018-176348	12/2017	JP	N/A
2019-094210	12/2018	JP	N/A
2019-218165	12/2018	JP	N/A
2019-218166	12/2018	JP	N/A
2020-001844	12/2019	JP	N/A
2020-040788	12/2019	JP	N/A
2021-041468	12/2020	JP	N/A
WO 2016/033172	12/2015	WO	N/A
WO 2019/049772	12/2018	WO	N/A
WO 2019/244846	12/2018	WO	N/A

OTHER PUBLICATIONS

Japanese Office Action issued Sep. 27, 2024 in Japanese Application 2021-145766, (with unedited computer-generated English translation), 9 pages. cited by applicant

Combined Chinese Office Action and Search Report issued Jul. 12, 2024 in Chinese Patent Application No. 202210041720.7, (with English translation), 8 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-004521, filed on Jan. 14, 2021; and Japanese Patent Application No. 2021-145766, filed on Sep. 7, 2021; the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a holding device, a cargo handling apparatus, and a holding method.

BACKGROUND

(3) There is an apparatus that holds an article. It is desirable to improve the convenience of such an apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a holding device according to a first embodiment;

(2) FIG. 2 is a side view illustrating a portion of the holding device according to the first embodiment;

(3) FIG. 3 is a perspective view illustrating a portion of the holding device according to the embodiment;

(4) FIG. 4 is a perspective view illustrating a portion of the holding device according to the embodiment;

(5) FIGS. 5A and 5B are side views illustrating a portion of the holding device according to the first embodiment;

(6) FIGS. 6A to 6C are schematic views illustrating the first operation of the holding device according to the first embodiment;

(7) FIGS. 7A to 7C are schematic views illustrating the first operation of the holding device according to the first embodiment;

(8) FIGS. 8A and 8B are schematic views illustrating the first operation of the holding device according to the first embodiment;

(9) FIGS. 9A and 9B are schematic views illustrating the second operation of the holding device according to the first embodiment;

(10) FIGS. 10A and 10B are schematic views illustrating the second operation of the holding device according to the first embodiment;

(11) FIGS. 11A and 11B are schematic views illustrating the second operation of the holding device according to the first embodiment;

(12) FIGS. 12A and 12B are schematic views illustrating the second operation of the holding device according to the first embodiment;

(13) FIG. 13 is a perspective view illustrating a portion of a holding device according to a first modification of the first embodiment;

(14) FIGS. 14A and 14B are perspective views illustrating the first and second holders of a holding device according to a second modification of the first embodiment;

(15) FIG. 15 is a perspective view illustrating a portion of a holding device according to a second modification of the first embodiment;

(16) FIGS. 16A to 16C are side views schematically illustrating a first holder and a second holder of a holding device according to a third modification of the embodiment;

(17) FIGS. 17A to 17C are side views schematically illustrating a first holder and a second holder of a holding device according to a fourth modification of the embodiment;

(18) FIG. 18 is a side view schematically illustrating a first holder and a second holder of a holding device according to a fifth modification of the embodiment;

(19) FIGS. 19A to 19C are side views schematically illustrating an operation of the holding device according to the fifth modification of the embodiment;

(20) FIGS. 20A to 20C are side views schematically illustrating a first holder and a second holder of a holding device according to a sixth modification of the embodiment;

(21) FIG. 21 is a side view schematically illustrating a portion of a holding device according to a seventh modification of the embodiment;

(22) FIGS. 22A and 22B are side views schematically showing operations of the holding device according to the seventh modification of the embodiment;

(23) FIGS. 23A to 23D are side views schematically showing states when the suction hand according to the embodiment holds the article;

(24) FIG. 24 is a front view schematically illustrating a portion of a holding device according to an eighth modification of the embodiment;

(25) FIG. 25 is a front view schematically showing a state when the holding device according to the eighth modification of the embodiment holds articles;

(26) FIG. 26 is a front view schematically showing a state when the holding device according to the eighth modification of the embodiment holds articles; and

(27) FIG. 27 is a schematic view illustrating a hardware configuration.

DETAILED DESCRIPTION

(28) According to one embodiment, a holding device includes a first holder configured to hold an upper surface of an article, and a second holder configured to hold a side surface of the article. The holding device performs at least a first operation of the first and second holders holding the article, and a second operation of only the first holder holding the article.

(29) Various embodiments are described below with reference to the accompanying drawings.

(30) The drawings are schematic and conceptual; and the relationships between the thickness and width of portions, the proportions of sizes among portions, etc., are not necessarily the same as the actual values. The dimensions and proportions may be illustrated differently among drawings, even for identical portions.

(31) In the specification and drawings, components similar to those described previously or illustrated in an antecedent drawing are marked with like reference numerals, and a detailed description is omitted as appropriate.

First Embodiment

(32) FIG. 1 is a perspective view illustrating a holding device according to a first embodiment.

(33) As illustrated in FIG. 1, the holding device **100** according to the first embodiment includes a first holder **1**, a second holder **2**, drivers **11x**, **11y**, **12x**, **12y**, and **13z**, base plates **20a** and **20b**, a frame **21**, a lift **30**, and a controller **50**.

(34) The first holder **1** is configured to hold an upper surface of an article. The second holder **2** is configured to hold a side surface of the article. The article can be more stably held by both the first holder **1** and the second holder **2** holding the article. Or, even when another article is frontward of the article, only the article can be selectively held by only the first holder **1** holding the upper surface of the article.

(35) Here, an X-direction (a longitudinal direction), a Y-direction (a lateral direction), and a Z-direction (a vertical direction) are used in the description. The Z-direction is parallel to a direction connecting the first holder **1** and the article when the first holder **1** holds the article. The X-direction is parallel to a direction connecting the second holder **2** and the article when the second holder **2** holds the article. The Z-direction crosses the X-direction. The Y-direction crosses a plane parallel to the X-direction and the Z-direction. For example, the X-direction and the Y-direction are parallel to a horizontal plane. The Z-direction is parallel to the vertical direction. The X-direction, the Y-direction, and the Z-direction are orthogonal to each other.

(36) FIG. 2 is a side view illustrating a portion of the holding device according to the first embodiment.

(37) In the example of FIGS. 1 and 2, the first holder **1** and the second holder **2** hold (stably grip) an article by vacuum-gripping.

(38) As illustrated in FIG. 2, the first holder **1** includes a housing **1a**, and a gripping unit **1b** that is mounted to the housing **1a**. The housing **1a** includes piping, valves, etc., necessary for vacuum-gripping. The gripping unit **1b** includes a rod **1c** and a suction pad **1d**. The rod **1c** extends in the Z-direction. The suction pad **1d** is located at the tip of the rod **1c**. The gripping surface of the tip of the suction pad **1d** faces downward.

(39) Similarly to the first holder **1**, the second holder **2** includes a housing **2a**, and a gripping unit **2b** that is mounted to the housing **2a**. The housing **2a** includes piping, valves, etc., necessary for vacuum-gripping. The gripping unit **2b** includes a rod **2c** and a suction pad **2d**. The rod **2c** extends in the X-direction when the second holder **2** holds the article. The suction pad **2d** is located at the tip of the rod **2c**. The gripping surface of the tip of the suction pad **2d** faces frontward when the second holder **2** holds the article.

(40) The internal spaces of the gripping units **1b** and **2b** are connected to piping and valves located inside the housings **1a** and **2a** and to a not-illustrated exhaust system such as an exhaust device located outside the first holder **1** and the second holder **2**, etc. The internal space of the gripping

unit **1b** can be exhausted independently of the internal space of the gripping unit **2b**.

(41) Pluralities of the gripping units **1b** and **2b** may be included. In the example of FIGS. **1** and **2**, the first holder **1** includes multiple gripping units **1b** arranged in two mutually-orthogonal directions. The second holder **2** includes multiple gripping units **2b** arranged in one direction.

(42) FIG. **3** is a perspective view illustrating a portion of the holding device according to the embodiment.

(43) A driver **11x** (an example of a first driver) moves the first holder **1** in the X-direction. A driver **11y** moves the first holder **1** in the Y-direction.

(44) For example, as illustrated in FIG. **3**, the driver **11x** includes an actuator **11a**, a pulley **11b**, a belt **11c**, a bar **11d**, and a guide **11e**. In the example, the actuator **11a** is a motor. The rotary shaft of the actuator **11a** is parallel to the Y-direction. The rotary shaft of the actuator **11a** is coupled to the pulley **11b**. Another pulley **11b** is positioned distant to the pulley **11b** in the X-direction. The belt **11c** is threaded over the pair of pulleys **11b**. The bar **11d** is fixed with respect to the belt **11c**. The bar **11d** engages the guide **11e** that extends along the X-direction. The belt **11c** is moved along the X-direction between the pair of pulleys **11b** when the actuator **11a** rotates the pulley **11b** around the Y-direction. The bar **11d** that is fixed with respect to the belt **11c** moves in the X-direction along the guide **11e**.

(45) A rod **11f** extends along the Y-direction and is fixed with respect to the bar **11d**. In the example, a pair of rods **11f** is separated from each other in the X-direction. The first holder **1** is coupled to each of the rods **11f**. The rods **11f** and the first holder **1** move in the X-direction according to the movement of the bar **11d**.

(46) The driver **11y** moves in the X-direction together with the bar **11d**. The driver **11y** includes an actuator **11h**, a pulley **11i**, and a belt **11j**. The actuator **11h** is a motor. The rotary shaft of the actuator **11h** is coupled to the pulley **11i**. Another pulley **11i** is positioned distant to the pulley **11i** in the Y-direction. The belt **11j** is threaded over the pair of pulleys **11i**.

(47) The first holder **1** is slidably coupled to the rod **11f** in the Y-direction. The first holder **1** is fixed with respect to the belt **11j**. The belt **11j** is moved along the Y-direction between the pair of pulleys **11i** when the actuator **11h** rotates the pulley **11i**. The first holder **1** that is fixed with respect to the belt **11j** moves in the Y-direction.

(48) A driver **12x** (an example of a second driver) moves the second holder **2** in the X-direction. A driver **12y** moves the second holder **2** in the Y-direction.

(49) FIG. **4** is a perspective view illustrating a portion of the holding device according to the embodiment.

(50) For example, as illustrated in FIGS. **3** and **4**, the driver **12x** is located higher than the driver **11x**, and includes an actuator **12a**, a pulley **12b**, a belt **12c**, a bar **12d**, and a guide **12e**. In the example, the actuator **12a** is a motor. The rotary shaft of the actuator **12a** is parallel to the Y-direction. The rotary shaft of the actuator **12a** is coupled to the pulley **12b**. Another pulley **12b** is positioned distant to the pulley **12b** in the X-direction. The belt **12c** is threaded over the pair of pulleys **12b**. The bar **12d** is fixed with respect to the belt **12c**. The bar **12d** engages the guide **12e** that extends along the X-direction. The belt **12c** is moved along the X-direction between the pair of pulleys **12b** when the actuator **12a** rotates the pulley **12b** around the Y-direction. The bar **12d** that is fixed with respect to the belt **12c** moves in the X-direction along the guide **12e**.

(51) One end of a link **12k** is rotatably coupled to the bar **12d**. A bar **12m** is rotatably coupled to the other end of the link **12k**. A rod **12f** extends along the Y-direction and is fixed with respect to the bar **12m**. In the example, a pair of rods **12f** that is separated from each other in the X-direction is fixed with respect to the bar **12m**. Each of the pair of links **12k** that is separated from each other in the X-direction is rotatably coupled to the bar **12m**. The link **12k**, the bar **12m**, the rod **12f**, and the second holder **2** move in the X-direction according to the movement of the bar **11d**.

(52) The driver **12y** moves in the X-direction together with the bar **12d**. The driver **12y** includes an actuator **12h** (an example of a fourth driver), a pulley **12i**, and a belt **12j**. The actuator **12h** is a

motor. The rotary shaft of the actuator **12h** is coupled to the pulley **12i**. Another pulley **12i** is positioned distant to the pulley **12i** in the Y-direction. The belt **12j** is threaded over the pair of pulleys **12i**. The rotary shaft of the actuator **12h** is parallel to the X-direction and is coupled to one of the pulleys **12i**.

(53) The second holder **2** is movably coupled to the rod **12f** in the Y-direction. The second holder **2** is fixed with respect to the belt **12j**. The belt **12j** is moved along the Y-direction between the pair of pulleys **12i** when the actuator **12h** rotates the pulley **12i**. The second holder **2** that is fixed with respect to the belt **12j** moves in the Y-direction.

(54) An actuator **12n** is fixed with respect to one of the bars **12m**. For example, the actuator **12n** is a motor. The rotary shaft of the actuator **12n** is parallel to the Y-direction and coupled to the other end of one of the links **12k**. The link **12k** is rotated with respect to the bar **12m** by the driving of the actuator **12n**. Thereby, the second holder **2** is rotated around the one end of the link **12k**. The X-direction position and the Z-direction position of the second holder **2** are changed by the rotation of the second holder **2**. For example, the second holder **2** rotates while maintaining the orientations of the suction pad **2d** and the rod **2c**.

(55) The one end of the link **12k** is positioned higher than the first holder **1**. In other words, the rotation center of the second holder **2** is positioned higher than the first holder **1**. The second holder **2** is switched between a state in which the second holder **2** is higher than the first holder **1** and a state in which the second holder **2** is lower than the first holder **1** by rotating the second holder **2** around its rotation center.

(56) The base plates **20a** and **20b** are supporters for supporting the first holder **1** and the second holder **2**. The base plate **20a** (a first supporter) and the base plate **20b** (a second supporter) are plate members that spread along the X-Z plane and are separated from each other in the Y-direction. The pulleys **11b** and **12b** are mounted to the base plate **20a**. The base plates **20a** and **20b** support the first holder **1** via the pulley **11b**, the belt **11c**, the bar **11d**, and the rod **11f**. Also, the base plates **20a** and **20b** support the second holder **2** via the pulley **12b**, the belt **12c**, the bar **12d**, the link **12k**, the bar **12m**, and the rod **12f**.

(57) The pulley **11b**, the belt **11c**, the bar **11d**, the guide **11e**, the pulley **12b**, the belt **12c**, the bar **12d**, the guide **12e**, the link **12k**, and the bar **12m** are mounted to each of the base plates **20a** and **20b** and are positioned between the base plates **20a** and **20b** when viewed along the Z-direction. The rod **11f** is fixed to the bars **11d** that face each other in the Y-direction. The rod **12f** is fixed to the bars **12d** that face each other in the Y-direction. The actuators **11a** and **12a** are mounted to the base plate **20a**. The actuators **11a** and **12a** may be mounted to the base plate **20b** or to both the base plates **20a** and **20b**.

(58) The pulley **11b** that is at the front of the base plate **20a** and the pulley **11b** that is at the front of the base plate **20b** are coupled to each other by a rod **20c** that extends in the Y-direction. The drive force of the actuator **11a** mounted to the base plate **20a** is transferred to the pulley **11b** of the base plate **20b** by the rod **20c**. The pair of pulleys **11b** of the base plate **20b** rotate synchronously with the pair of pulleys **11b** of the base plate **20a**. Thereby, the bar **11d** of the base plate **20b** is moved in the X-direction synchronously with the bar **11d** of the base plate **20a**.

(59) The pulley **12b** that is at the front of the base plate **20a** and the pulley **12b** that is at the front of the base plate **20b** are coupled to each other by a rod **20d** that extends in the Y-direction. The drive force of the actuator **12a** mounted to the base plate **20a** is transferred to the pulley **12b** of the base plate **20b** by the rod **20d**. The pair of pulleys **12b** of the base plate **20b** rotates synchronously with the pair of pulleys **12b** of the base plate **20a**. Thereby, the bar **12m** of the base plate **20b** is moved in the X-direction synchronously with the bar **12m** of the base plate **20a**.

(60) The first holder **1** and the second holder **2** can be more stably supported by the rods **11f** and **12f** and the rods **20c** and **20d** bridging between the base plates **20a** and **20b**. For example, the positions in the Z-direction of the first and second holders **1** and **2** can be more stable.

(61) As illustrated in FIG. 1, multiple drivers **13z** (examples of the third driver) are arranged in the

Y-direction. A pair of drivers **13z** moves the base plates **20a** and **20b** in the Z-direction. The first holder **1** and the second holder **2** are moved in the Z-direction by the operation of the drivers **13z**.

(62) For example, each of the drivers **13z** includes an actuator **13a**, a pair of pulleys **13b**, a belt **13c**, a block **13d**, and a guide **13e**. The actuator **13a** is a motor. The rotary shaft of the actuator **13a** is parallel to the Y-direction. The rotary shaft of the actuator **13a** is coupled to one of the pulleys **13b**. The pair of pulleys **13b** is separated from each other in the Z-direction. The belt **13c** is threaded over the pair of pulleys **13b**.

(63) The blocks **13d** are fixed respectively to the side surface of the base plate **20a** and the side surface of the base plate **20b**. The block **13d** is fixed to the belt **13c**. The base plates **20a** and **20b** respectively engage the guides **13e**. By the rotation of the actuator **13a**, the pulley **13b** is rotated around the Y-direction, and the belt **13c** is moved along the Z-direction between the pair of pulleys **13b**. Thereby, the base plates **20a** and **20b** that are fixed with respect to the belts **13c** are moved in the Z-direction.

(64) The drivers **13z** are mounted to the frame **21**. The first holder **1** and the second holder **2** are movable in the X-direction, the Y-direction, and the Z-direction with respect to the frame **21** by the drivers **11x**, **11y**, **12x**, **12y**, and **13z**.

(65) For example, the frame **21** includes vertical frames **21a** to **21d** and horizontal frames **21e** to **21k**. The vertical frames **21a** and **21b** are separated from each other in the X-direction. The vertical frames **21c** and **21d** are separated from each other in the X-direction. The vertical frames **21a** and **21b** face the vertical frames **21c** and **21d** in the Y-direction. The horizontal frames **21e** and **21f** couple the lower ends of the vertical frames **21a** and **21b** to each other and couple the upper ends of the vertical frames **21a** and **21b** to each other. The horizontal frames **21g** and **21h** couple the lower ends of the vertical frames **21c** and **21d** to each other and couple the upper ends of the vertical frames **21c** and **21d** to each other. The horizontal frames **21i** and **21j** couple the lower ends of the vertical frames **21b** and **21d** to each other and couple the upper ends of the vertical frames **21b** and **21d** to each other. The horizontal frame **21k** couples the upper ends of the vertical frames **21a** and **21c** to each other. The guides **13e** are mounted respectively to the vertical frames **21a** and **21c**.

(66) The lift **30** is located inside the frame **21**. In other words, the lift **30** is positioned inside a rectangular region surrounded with the vertical frames **21a** to **21d** when viewed along the Z-direction. The upper surface of the lift **30** is parallel to the X-direction and the Y-direction. The lift **30** is moved in the Z-direction by a not-illustrated driver.

(67) As shown in FIG. 1, the lift **30** includes a pair of rollers **30a** and a belt **30b**. The pair of rollers **30a** are separated from each other in the X-direction. The belt **30b** is threaded over the pair of rollers **30a**. The rotary shafts of the rollers **30a** are parallel to the Y-direction. The belt **30b** is moved along the X-direction between the pair of rollers **30a** when a not-illustrated motor rotates the rollers **30a**.

(68) A conveyor **C** is located outside the frame **21**. A driver of the lift **30** sets the lift **30** to a state in which the lift **30** is at the same level as the conveyor **C**, a state in which the lift **30** is at a lower position than the conveyor **C**, or a state in which the lift **30** is at a higher position than the conveyor **C**.

(69) When a pallet **P** on which articles are loaded is placed in front of the holding device **100**, the holding device **100** sequentially transfers the articles to the lift **30**. The lift **30** is set to the same level as the conveyor **C**. Subsequently, the articles that are placed on the lift **30** are transferred onto the conveyor **C**. In other words, the holding device **100** functions as a cargo handling apparatus **100a** that uses the first holder **1**, the second holder **2**, the lift **30**, etc., to unload, onto the conveyor **C**, the articles loaded on the pallet **P**.

(70) The holding device **100** may include a detector **35**. The detector **35** detects the articles that are loaded on the pallet **P**. For example, the detector **35** detects the position of an article **A**. The detector **35** includes at least one selected from the group consisting of an image sensor, a through-beam sensor, a laser sensor, and an infrared sensor. Favorably, the detector **35** is a camera that

includes an image sensor.

(71) The controller **50** is connected with and controls the drivers, the exhaust device, and the detector **35** of the holding device **100**. The controller **50** processes data necessary for holding and transferring the articles A. The controller **50** includes a processing circuit that includes a central processing unit, memory in which programs are stored, etc. The controller **50** may calculate the positions of the articles based on an image acquired by the detector **35**.

(72) FIGS. **5A** and **5B** are side views illustrating a portion of the holding device according to the first embodiment.

(73) The controller **50** switches between a first state in which the second holder **2** is at a first position and a second state in which the second holder **2** is at a second position.

(74) The second holder **2** can hold the side surface of the article at the first position. The second holder **2** is positioned lower than at least a portion of the first holder **1** when the second holder **2** is at the first position. For example, the gripping unit **2b** is positioned lower than the housing **1a** and the upper end of the rod **1c** of the first holder **1**.

(75) The second position is separated from the first position. For example, the second position is higher than the first position. The second holder **2** is positioned higher than at least a portion of the first holder **1** when the second holder **2** is at the second position. For example, the gripping unit **2b** is positioned higher than the housing **1a** and the gripping unit **1b** of the first holder **1**. When the actuator **12n** operates, the link **12k** is rotated, and the position of the second holder **2** is changed. The second holder **2** is movable between the first position and the second position by the actuator **12n**.

(76) FIG. **5A** illustrates the state when the second holder **2** is at the first position. FIG. **5B** illustrates the state when the second holder **2** is at the second position. As illustrated in FIG. **5A**, at least a portion of the second holder **2** is at the same level as the first holder **1** when the second holder **2** is at the first position. As illustrated in FIG. **5B**, the second holder **2** is positioned higher than the first holder **1** when the second holder **2** is at the second position. In the example, at least a portion of the second holder **2** is positioned directly above the first holder **1**.

(77) The holding device **100** performs the first operation when transferring articles of a first arrangement. The position of the second holder **2** is set to the first position in the first operation. The holding device **100** performs the second operation when transferring articles of a second arrangement. The position of the second holder **2** is set to the second position in the second operation. In the first arrangement, articles of substantially the same size are arranged in the X-direction and the Y-direction and loaded in the Z-direction. In the second arrangement, articles of different sizes are irregularly loaded.

(78) FIGS. **6A** to **6C**, FIGS. **7A** to **7C**, FIG. **8A**, and FIG. **8B** are schematic views illustrating the first operation of the holding device according to the first embodiment.

(79) FIG. **9A**, FIG. **9B**, FIG. **10A**, FIG. **10B**, FIG. **11A**, FIG. **11B**, FIG. **12A**, and FIG. **12B** are schematic views illustrating the second operation of the holding device according to the first embodiment.

(80) When transferring articles of the first arrangement, the holding device **100** sequentially transfers from the article positioned at the front. The controller **50** drives the actuator **12n** to position the second holder **2** at the first position as illustrated in FIG. **6A**. The controller **50** positions the upper surface of the lift **30** at the same level as the bottom surface of the article A to be transferred.

(81) The controller **50** drives the drivers **11x** and **11y** to move the first holder **1** directly above the article A as illustrated in FIG. **6B**. The controller **50** drives the driver **13z** to press the first holder **1** onto the upper surface of the article A as illustrated in FIG. **6C**. Each rod **1c** is displaced, and the suction pad **1d** deforms according to the shape of the upper surface. The controller **50** operates the exhaust system to set the internal pressure of the gripping unit **1b** to be less than atmospheric pressure. Thereby, the first holder **1** holds the upper surface of the article A.

(82) The controller **50** drives the drivers **12x** and **12y** to press the second holder **2** onto the side surface of the article **A** as illustrated in FIG. **7A**. Each rod **2c** is displaced, and the suction pad **2d** deforms according to the shape of the side surface. The controller **50** operates the exhaust system to set the internal pressure of the gripping unit **2b** to be less than atmospheric pressure. Thereby, the second holder **2** holds the side surface of the article **A**.

(83) As illustrated in FIG. **7B**, the controller **50** operates the drivers **11x** and **12x** to transfer the article **A** from the pallet **P** onto the lift **30**. At this time, the controller **50** may operate the driver **13z** to lift the article **A**. When the article **A** is transferred onto the lift **30**, the controller **50** releases the holding by the first and second holders **1** and **2**. Namely, the controller **50** increases the pressure of the internal space of the gripping unit **1b** and the pressure of the internal space of the gripping unit **2b** and weakens the gripping force of the first and second holders **1** and **2**. The controller **50** operates the drivers **12x** and **13z** to separate the first and second holders **1** and **2** from the article **A** as illustrated in FIG. **7C**.

(84) As illustrated in FIG. **8A**, the controller **50** positions the upper surface of the lift **30** at the same level as the upper surface of the conveyor **C**. As illustrated in FIG. **8B**, the lift **30** is operated to transfer the article **A** onto the conveyor **C**. The conveyor **C** transfers the article **A** to another location. The holding device **100** repeats the first operation illustrated in FIGS. **6A** to **8B** until all of the articles **A** placed on the pallet **P** are transferred onto the conveyor **C**.

(85) When transferring articles of the second arrangement, the holding device **100** sequentially transfers from an article that is at a high position. As illustrated in FIG. **9A**, the controller **50** drives the actuator **12n** to position the second holder **2** at the second position. At this time, the controller **50** may position the upper surface of the lift **30** at the same level as the bottom surface of the article **A** to be transferred. The controller **50** drives the drivers **11x** and **11y** to move the first holder **1** directly above the article **A** that is the transfer object as illustrated in FIG. **9B**.

(86) The transfer object is, for example, the article **A** of which the upper surface is at the highest position. When the first holder **1** approaches an article **A** of which the upper surface is at a low position, a portion of the holding device **100** interferes with the articles **A** of which the upper surfaces are at higher positions, and there is a possibility that the article **A** cannot be transferred. The transfer object may be determined by a user of the holding device **100**. The positions of the upper surfaces of the articles **A** may be calculated based on the detection result of the detector **35**. The controller **50** determines the article **A** to be transferred based on the calculated positions of the upper surfaces.

(87) The controller **50** drives the driver **13z** to press the first holder **1** onto the upper surface of the article **A** as illustrated in FIG. **10A**. The controller **50** operates the exhaust system to set the internal pressure of the gripping unit **1b** to be less than atmospheric pressure. Thereby, the first holder **1** holds the upper surface of the article **A**. The controller **50** operates the driver **13z** to lift the article **A** as illustrated in FIG. **10B**. At this time, the second holder **2** is still at the second position. The second holder **2** does not hold the article **A**.

(88) The controller **50** operates the drivers **11x** and **12x** to transfer the article **A** from the pallet **P** onto the lift **30**. The controller **50** operates the driver **13z** to place the article **A** onto the lift **30** as illustrated in FIG. **11A**. The controller **50** releases the holding by the first holder **1**. The controller **50** operates the driver **13z** to separate the first and second holders **1** and **2** from the article **A** as illustrated in FIG. **11B**.

(89) As illustrated in FIG. **12A**, the controller **50** positions the upper surface of the lift **30** at the same level as the upper surface of the conveyor **C**. As illustrated in FIG. **12B**, the lift **30** is operated to transfer the article **A** onto the conveyor **C**. The conveyor **C** transfers the article **A** to another location. The holding device **100** repeats the second operation illustrated in FIGS. **9A** to **12B** until all of the articles **A** placed on the pallet **P** are transferred onto the conveyor **C**.

(90) Data of the arrangement of the articles **A** on the pallet **P** may be input by the user or may be transmitted to the controller **50** by a communicator. For example, the controller **50** receives the data

of the arrangement. The controller **50** switches the position of the second holder **2** based on the received data.

(91) Advantages of the first embodiment will now be described.

(92) When the multiple articles **A** are loaded in the regular first arrangement, the heights of the articles **A** are the same as those illustrated in FIG. **6A**. For example, products of the same product type are packaged in each article **A**. In such a case, the article **A** can be more stably held by both the first holder **1** and the second holder **2** by transferring the article **A** at the front first. Compared to when the article **A** is held by only one of the first holder **1** or the second holder **2**, the application of a locally-large force to the article **A** can be suppressed. The likelihood of damaging the article **A** can be reduced thereby. Both the first holder **1** and the second holder **2** are used for the articles **A** of the first arrangement.

(93) When the multiple articles **A** are loaded at the irregular second arrangement, the sizes of the articles **A** are different from each other as illustrated in FIG. **9A**. For example, the articles **A** are courier parcels. In such a case, when the first holder **1** approaches an article **A** of which the upper surface is at a low position, a portion of the holding device **100** interferes with other articles **A** of which the upper surfaces are at higher positions, and there is a possibility that the other articles **A** may be damaged. Therefore, the holding device **100** holds the article **A** of which the upper surface is at the highest position when transferring. For the articles **A** of the second arrangement, only the first holder **1** is used, and the second holder **2** is not used.

(94) Conventionally, different holding devices are used between when transferring the articles **A** of the first arrangement illustrated in FIG. **6A** and when transferring the articles **A** of the second arrangement illustrated in FIG. **9A**. In other words, a holding device that includes the first and second holders **1** and **2** is used for the articles **A** of the first arrangement. A holding device that includes only the first holder **1** is used for the articles **A** of the second arrangement. However, in a conventional method, it is necessary to assign the pallets to the holding devices according to the type of the holding device and the arrangement of the articles. Therefore, management is complex. Also, the operation rate of the holding device may decrease.

(95) For this problem, the holding device **100** according to the first embodiment performs the first operation of holding the article **A** with both the first holder **1** and the second holder **2**, and the second operation of holding the article **A** with only the first holder **1**. By selectively performing one of the first operation or the second operation, the article **A** can be held regardless of the arrangement. According to the first embodiment, the convenience of the holding device **100** can be improved. For example, the operation rate of the holding device **100** can be improved thereby.

(96) Favorably, the holding device **100** switches between the first state in which the second holder **2** is at the first position and the second state in which the second holder **2** is at the second position. When holding the articles **A** of the second arrangement, there is a possibility that the second holder **2** may interfere with an article **A** other than the holding object. When the second holder **2** is at the second position in the second operation, the likelihood of the second holder **2** interfering with the other article **A** can be reduced.

(97) More favorably, the second position is higher than the first holder **1**. The likelihood of the second holder **2** interfering with the other article **A** in the second operation can be further reduced thereby.

(98) More favorably, at least a portion of the second holder **2** is positioned directly above the first holder **1** when the second holder **2** is at the second position. The likelihood of the second holder **2** interfering with the other article **A** in the second operation can be further reduced thereby.

(99) Favorably, the distance in the Y-direction between the base plate **20a** and the base plate **20b** is greater than the length in the Y-direction of the lift **30**. The lift **30** is positioned between the base plate **20a** and the base plate **20b** when viewed along the Z-direction. Therefore, the interference of the lift **30** with the base plates **20a** and **20b** when the base plates **20a** and **20b** are moved in the Z-direction or when the lift **30** is moved in the Z-direction can be suppressed.

(100) The first holder **1** and the second holder **2** are movable to a position that does not interfere with the lift **30** when the lift **30** moves to be higher than the first holder **1**. The interference of the lift **30** with the first holder **1**, the second holder **2**, the base plate **20a**, and the base plate **20b** can be suppressed thereby, and the operation rate of the holding device **100** can be improved.

(101) Specifically, in the holding device **100**, the first holder **1** and the second holder **2** are movable further frontward than the lift **30**. The first holder **1** and the second holder **2** do not overlap the lift **30** when viewed along the Z-direction. For example, when only the first holder **1** holds the article **A** as illustrated in FIGS. **9B** to **10B**, the first holder **1** and the second holder **2** can be moved in the Z-direction regardless of the Z-direction position of the lift **30**.

(102) First Modification

(103) In the holding device according to the first embodiment, the second holder **2** may be fixed with respect to the bar **12d**. For example, the two ends of the link **12k** are fixed respectively to the bars **12d** and **12m**. The actuator **12n** is not included.

(104) FIG. **13** is a perspective view illustrating a portion of a holding device according to a first modification of the first embodiment.

(105) Parts of the holding device **110** such as the bar **12d**, the guide **12e**, the pulley **12i**, the link **12k**, etc., are not illustrated in FIG. **13**. In the holding device **100**, the second holder **2** is movable by the actuator **12n** in the Z-direction with respect to the first holder **1**. Conversely, in the holding device **110** according to the first modification, the second holder **2** is movable only in the X-direction or the Y-direction with respect to the first holder **1**.

(106) The controller **50** switches between the first state in which the second holder **2** is at the first position and the second state in which the second holder **2** is at the second position. The second holder **2** can hold the side surface of the article at the first position. The second position is further backward of the first position and is at the same level as the first position. For example, the second position is directly above the lift **30**.

(107) When the first operation is performed, the first holder **1** moves to the first position at the front. When the second operation is performed, the second holder **2** withdraws to the second position at the back as illustrated in FIG. **13**. According to the first modification, similarly to the holding device **100**, the article **A** can be held regardless of the arrangement. Therefore, the convenience of the holding device **110** can be improved.

(108) Second Modification

(109) FIGS. **14A** and **14B** are perspective views illustrating the first and second holders of a holding device according to a second modification of the first embodiment.

(110) In the holding device according to the first embodiment, the second holder **2** may be coupled to the first holder **1**. In the holding device **120** according to the second modification, the second holder **2** is detachably coupled to the first holder **1**. FIG. **14A** illustrates a state when the second holder **2** is coupled to the first holder **1**. FIG. **14B** illustrates a state when the second holder **2** is separated from the first holder **1**.

(111) A coupler **15** is included in the first and second holders **1** and **2**. The coupler **15** couples the first holder **1** and the second holder **2**. The coupler **15** includes a first coupling part **15a** and a second coupling part **15b**. The first coupling part **15a** and the second coupling part **15b** are mounted respectively to the first and second holders **1** and **2**. In the example, the first coupling part **15a** and the second coupling part **15b** are coupled by a magnetic force. The first coupling part **15a** and the second coupling part **15b** may be mechanically coupled.

(112) The first coupling part **15a** includes an electromagnet. A current is supplied to the first coupling part **15a** via the first holder **1**. A magnetic field is generated at the periphery of the first coupling part **15a** when electrical power is supplied to the first coupling part **15a**. Thereby, as illustrated in FIG. **14A**, the first coupling part **15a** and the second coupling part **15b** are coupled by a gripping force.

(113) The second coupling part **15b** is mounted to the second holder **2** via an actuator **15c** (an

example of a second driver). In the example, the actuator **15c** is a power cylinder. The actuator **15c** extends and retracts along the X-direction. The actuator **15c** functions as a driver that moves the second holder **2** in the X-direction with respect to the first holder **1**.

(114) The first coupling part **15a** and the second coupling part **15b** are separable when the supply of the current to the first coupling part **15a** is stopped. The second holder **2** is separated from the first holder **1** by moving the first holder **1** in the state in which the coupling is released.

(115) FIG. **15** is a perspective view illustrating a portion of a holding device according to a second modification of the first embodiment.

(116) For example, the holding device **120** according to the second modification includes a placement platform **40**. The placement platform **40** is located at a position that does not interfere with the first and second holders **1** and **2** when the holding device **120** transfers the article. The placement platform **40** is located at a position that does not interfere with the lift **30** when the lift **30** moves in the Z-direction. For example, the placement platform **40** is located higher than the range of movement of the lift **30** in the Z-direction.

(117) When performing the first operation, the second holder **2** is coupled to the first holder **1**. When performing the second operation, the second holder **2** is placed on the placement platform **40**. In other words, the state in which the second holder **2** is coupled to the first holder **1** corresponds to the first state. The state in which the second holder **2** is placed on the placement platform **40** corresponds to the second state. The second position is on the placement platform **40**. The first holder **1** holds the article at a lower position than the second position.

(118) When the second operation is performed after performing the first operation, the drivers **11x** and **11y** move the first holder **1** and the second holder **2** so that the second holder **2** is positioned on the placement platform **40**. The second holder **2** is placed on the placement platform **40** by releasing the coupling of the coupler **15**. Subsequently, the second operation is performed using the first holder **1**.

(119) According to the second modification, similarly to the holding device **100**, the convenience of the holding device **120** can be improved. Furthermore, the interference of the second holder **2** with the article can be avoided by placing the second holder **2** on the placement platform **40** in the second operation. The operation rate of the holding device **120** can be improved by providing the placement platform **40** at a position that does not interfere with the other components (e.g., the lift **30**).

(120) Third Modification

(121) FIGS. **16A** to **16C** are side views schematically illustrating a first holder and a second holder of a holding device according to a third modification of the embodiment.

(122) In the holding device **130** according to the third modification, the second holder **2** is coupled to the first holder **1**. The driver **12x** moves the second holder **2** in the X-direction with respect to the first holder **1**. A driver **12z** moves the second holder **2** in the Z-direction with respect to the first holder **1**.

(123) For example, the second holder **2** is coupled to a coupling part **16a** via the driver **12x**. The coupling part **16a** is coupled to a support frame **16b** via the driver **12z**. The first holder **1** is fixed with respect to the support frame **16b**. The support frame **16b** is fixed with respect to an arm **16c**. For example, the arm **16c** is movable in the X-direction and the Y-direction by the drivers **11x** and **11y**. Or, the arm **16c** may be a portion of an X-Y orthogonal robot. The arm **16c** may be a portion of an arm of a vertical articulated robot or a horizontal articulated robot.

(124) FIG. **16A** illustrates the first state in which the second holder **2** is at the first position. When the second holder **2** moves from the first position to the second position, the driver **12x** moves the second holder **2** backward as illustrated in FIG. **16B**. Then, as illustrated in FIG. **16C**, the driver **12z** moves the second holder **2** upward. Thereby, the second holder **2** is positioned higher than at least a portion of the first holder **1** (the suction pad **1d**). FIG. **16C** illustrates the second state in which the second holder **2** is at the second position. The holding device **130** performs the first

operation of transferring the articles of the first arrangement in the first state illustrated in FIG.

16A. The holding device **130** performs the second operation of transferring the articles of the second arrangement in the second state illustrated in FIG. **16C**.

(125) When the second holder **2** is at the first position, the gripping unit **2b** is positioned lower than the housing **1a** and the upper end of the rod **1c** of the first holder **1**. When the second holder **2** is at the second position, the gripping unit **2b** is positioned higher than the lower end of the rod **1c** and the suction pad **1d** of the first holder **1**.

(126) According to the third modification, similarly to the holding device **100**, the convenience of the holding device **130** can be improved. By positioning the second holder **2** higher than at least a portion of the first holder **1** in the second operation, the interference of the second holder **2** with the article can be suppressed. The operation rate of the holding device **130** can be improved thereby.

(127) Fourth Modification

(128) FIGS. **17A** to **17C** are side views schematically illustrating a first holder and a second holder of a holding device according to a fourth modification of the embodiment.

(129) In the holding device **140** according to the fourth modification, the driver **12x** moves the second holder **2** in the X-direction with respect to the first holder **1**. A driver **11z** moves the first holder **1** in the Z-direction with respect to the second holder **2**.

(130) For example, the second holder **2** is coupled to the support frame **16b** via the driver **12x** and the coupling part **16a**. The first holder **1** is coupled to the support frame **16b** via the driver **11z**. The support frame **16b** is fixed with respect to the arm **16c**.

(131) FIG. **17A** illustrates the first state in which the second holder **2** is at the first position. When the holding device **140** transitions from the first state to the second state, the driver **12x** moves the second holder **2** backward as illustrated in FIG. **17B**. The driver **11z** moves the first holder **1** downward with respect to the second holder **2**. Thereby, as illustrated in FIG. **17C**, the second holder **2** is positioned higher than at least a portion of the first holder **1**. FIG. **17C** illustrates the second state in which the second holder **2** is at the second position. The holding device **140** performs the first operation in the first state illustrated in FIG. **17A**. The holding device **140** performs the second operation in the second state illustrated in FIG. **17C**.

(132) When the second holder **2** is at the first position, the gripping unit **2b** is positioned lower than the housing **1a** and the upper end of the rod **1c** of the first holder **1**. When the second holder **2** is at the second position, the gripping unit **2b** is positioned higher than the lower end of the rod **1c** and the suction pad **1d** of the first holder **1**.

(133) According to the fourth modification, similarly to the holding device **100**, the convenience of the holding device **140** can be improved. The interference of the second holder **2** with the article can be suppressed by positioning the second holder **2** higher than at least a portion of the first holder **1** in the second operation. The operation rate of the holding device **140** can be improved thereby.

(134) Fifth Modification

(135) FIG. **18** is a side view schematically illustrating a first holder and a second holder of a holding device according to a fifth modification of the embodiment.

(136) The tilts of the first and second holders **1** and **2** with respect to the X-Y plane may be changeable. For example, compared to the holding device **130**, the holding device **150** according to the fifth modification further includes a rod **16d**, a coupling part **16e**, and a driver **16z** as illustrated in FIG. **18**.

(137) The driver **16z** is mounted to the support frame **16b**. One end of the rod **16d** is coupled to the driver **16z**. The driver **16z** slides the rod **16d** in the Z-direction. The other end of the rod **16d** is coupled to the first holder **1** via the coupling part **16e**. The second holder **2** and the rod **16d** are rotatably coupled to the coupling part **16e** around the Y-direction.

(138) The first holder **1** is rotatable around the Y-direction with respect to the support frame **16b** around a rotary shaft **16f**. The Z-direction position of the front portion of the first holder **1** changes

according to the operation of the driver **16z**. The Z-direction position of the front portion of the first holder **1** is fixed. Therefore, the tilt of the first holder **1** with respect to the X-Y plane changes when the driver **16z** moves the rod **16d** in the Z-direction. Simultaneously, the tilt of the second holder **2** with respect to the X-Y plane also changes. The tilts of the first and second holders **1** and **2** with respect to the support frame **16b** and the arm **16c** are changed by the operation of the driver **16z**. (139) FIGS. **19A** to **19C** are side views schematically illustrating an operation of the holding device according to the fifth modification of the embodiment.

(140) FIG. **19A** illustrates a state when the holding device **150** is in the first state. FIG. **19B** illustrates a state when the holding device **150** is in the second state. FIG. **19C** illustrates the state when the holding device **150** is in the first state and the first holder **1** and the second holder **2** are tilted.

(141) The holding device **150** tilts the first holder **1** and the second holder **2** when holding the article or when placing the article. For example, the holding device **150** tilts the first holder **1** and the second holder **2** after placing the article and releasing the gripping of the article. Thereby, the contact area between the first holder **1** and the article and the contact area between the second holder **2** and the article can be reduced even when there is no space to move the second holder **2** backward. The friction between the second holder **2** and the article when raising the first holder **1** and the second holder **2** can be reduced. As a result, damage of the article can be suppressed, the load of the drivers raising the first holder **1** and the second holder **2** can be reduced, etc.

(142) Sixth Modification

(143) FIGS. **20A** to **20C** are side views schematically illustrating a first holder and a second holder of a holding device according to a sixth modification of the embodiment.

(144) The second holder **2** may rotate to change the orientation of the gripping surface of the suction pad **2d**. As illustrated in FIG. **20A**, the second holder **2** is coupled to the link **12k**. One end of the link **12k** is coupled to the actuator **12n**. The link **12k** and the second holder **2** are rotated by the actuator **12n**. The rotation center of the second holder **2** is positioned higher than the gripping surface of the suction pad **1d** and is parallel to the Y-direction. The level of the suction pad **2d** is changed by the rotation of the second holder **2**.

(145) FIG. **20A** illustrates the first state in which the second holder **2** is at the first position. When the second holder **2** is at the first position, the gripping unit **2b** is positioned lower than the upper end of the rod **1c**. When the second holder **2** moves from the first position to the second position, the second holder **2** is rotated by the actuator **12n** as illustrated in FIG. **20B**. The second holder **2** reaches the same level as the first holder **1**. In other words, the gripping surface of the suction pad **2d** reaches the same level as the gripping surface of the suction pad **1d**. The second holder **2** can hold the upper surface of the article when the first holder **1** holds the upper surface of the article.

(146) The second holder **2** may be slidable along the link **12k**. For example, a driver **12s** is coupled to the second holder **2** and the link **12k**. The driver **12s** causes the second holder **2** to slide along the link **12k**. As illustrated in FIG. **20C**, the distance between the first holder **1** and the second holder **2** is reduced by the operation of the driver **12s**. In other words, the distance between the second holder **2** and the article is reduced when the first holder **1** holds the article. Thereby, the second holder **2** can easily contact the upper surface of the article.

(147) FIG. **20B** or FIG. **20C** illustrate the second state in which the second holder **2** is at the second position. For example, the holding device **160** performs the first operation of transferring the articles of the first arrangement in the first state illustrated in FIG. **20A**. The holding device **160** performs the second operation of transferring the articles of the second arrangement in the second state illustrated in FIG. **20C**.

(148) According to the sixth modification, similarly to the holding device **100**, the convenience of the holding device **160** can be improved. Furthermore, the interference of the second holder **2** with the article can be suppressed by the second holder **2** approaching the first holder **1** in the second operation. For example, the interference between the second holder **2** and the lift **30** can be

suppressed, and the operation rate of the holding device **160** can be improved.

(149) Seventh Modification

(150) FIG. **21** is a side view schematically illustrating a portion of a holding device according to a seventh modification of the embodiment.

(151) As shown in FIG. **21**, the holding device **170** according to the seventh modification includes an arm **17a**, a base **17b**, a tilt mechanism **17c**, a linear guide **17d**, a coupling frame **17e**, a fixing mechanism **17f**, a linear guide **17g**, a coupling plate **17h**, a cylinder **17i**, a linkage **17k**, and a drive mechanism **17m** (an example of a fourth driver).

(152) The arm **17a** is connected with an X-Y orthogonal robot (an example of the first driver) that is included as the drivers **11x** and **11y**. The arm **17a** may be a portion of an X-Y orthogonal robot. The base **17b** is a plate member that spreads along the X-Z plane. The back end of the lower portion of the base **17b** is coupled with a tip **17a1** of the arm **17a** via a rotary shaft **17a2**, and is rotatable with respect to the tip **17a1**.

(153) The tilt mechanism **17c** is located between the arm **17a** and the base **17b**. Specifically, the tilt mechanism **17c** includes a cylinder **17c1**, a rotary shaft **17c2**, and a rotary shaft **17c3**. The arm **17a** includes a bent portion **17a3** that is bent downward. One end of the cylinder **17c1** is coupled to the bent portion **17a3** via the rotary shaft **17c2** and is rotatable with respect to the bent portion **17a3**. The other end of the cylinder **17c1** is coupled to the back end of the upper portion of the base **17b** via the rotary shaft **17c3** and is rotatable with respect to the back end. The rotation centers of the rotary shafts **17a2**, **17c2**, and **17c3** are parallel to the Y-direction.

(154) The cylinder **17c1** is extendable and retractable along the X-direction. The levels at which the cylinder **17c1**, the rotary shaft **17c2**, and the rotary shaft **17c3** are located are different from the level at which the rotary shaft **17a2** is located. Thereby, the angle of the base **17b** around the Y-direction is changed when the length of the cylinder **17c1** is changed. The cylinder **17c1**, the rotary shaft **17c2**, and the rotary shaft **17c3** are positioned above the rotary shaft **17a2** in the holding device **170**. For example, the front end of the base **17b** is tilted downward by the extension of the cylinder **17c1**.

(155) The linear guide **17d** is located at the side surface of the base **17b**. In the holding device **170**, two linear guides **17d** are located at one side surface of the base **17b**. The linear guide **17d** includes a rail and a block. The rail is fixed with respect to the base **17b** and is located along the Z-direction. The block is movable in the Z-direction along the rail. The coupling frame **17e** is fixed with respect to the block of the linear guide **17d**. Thereby, the coupling frame **17e** is movable in the Z-direction with respect to the base **17b**.

(156) The fixing mechanism **17f** is located at the side surface of the base **17b** proximate to the coupling frame **17e**. The fixing mechanism **17f** is switchable between a locked state in which the coupling frame **17e** is fixed and an unlocked state in which the coupling frame **17e** is movable. Specifically, the fixing mechanism **17f** includes a rod **17f1** and a clamber **17f2**. The rod **17f1** is fixed with respect to the base **17b** and extends along the Z-direction. The clamber **17f2** is coupled to the coupling frame **17e** and is slidable in the Z-direction along the rod **17f1**. In the locked state, the clamber **17f2** is fixed with respect to the rod **17f1**, and the coupling frame **17e** also is fixed thereby.

(157) The first holder **1** is fixed with respect to the coupling frame **17e**. The linear guide **17g** is located along the upper surface of the housing **1a**. The linear guide **17g** includes a rail and a block. The rail is fixed with respect to the housing **1a** and extends along the X-direction. The block is movable in the X-direction along the rail.

(158) The coupling plate **17h** is a plate member that extends along the X-direction. The front portion of the coupling plate **17h** is parallel to the X-Y plane and is fixed with respect to the blocks of the linear guides **17g**. Thereby, the coupling plate **17h** is movable in the X-direction with respect to the first holder **1**. The back portion of the coupling plate **17h** is slightly tilted downward.

(159) The cylinder **17i** (an example of the second driver) is coupled to the coupling plate **17h** and

the housing **1a** of the first holder **1**. The cylinder **17i** is extendable and retractable along the X-direction. The coupling plate **17h** moves in the X-direction with respect to the first holder **1** when the cylinder **17i** extends or contracts. Thereby, the second holder **2** that is coupled to the coupling plate **17h** is moved along the X-direction.

(160) The linkage **17k** is coupled between the coupling plate **17h** and the second holder **2**. The linkage **17k** includes a pair of links **17k1** and **17k2**. The links **17k1** and **17k2** extend parallel to each other and are synchronously rotatable. One end of the link **17k1** and one end of the link **17k2** are coupled respectively via rotary shafts **17k3** and **17k4** with the back end of the coupling plate **17h**. The rotary shafts **17k3** and **17k4** are rotatable with respect to the coupling plate **17h**. On the other hand, the links **17k1** and **17k2** are respectively fixed with respect to the rotary shafts **17k3** and **17k4**. Therefore, the links **17k1** and **17k2** also rotate when the rotary shafts **17k3** and **17k4** rotate with respect to the coupling plate **17h**.

(161) The drive mechanism **17m** is coupled to the coupling plate **17h** and the linkage **17k** and changes the Z-direction position of the second holder **2**. Specifically, the drive mechanism **17m** includes a cylinder **17m1**, a rotary shaft **17m2**, a rotary shaft **17m3**, and a bar **17m4**. The cylinder **17m1** is located along the back portion of the coupling plate **17h**. The tube main body of the cylinder **17m1** is coupled with the coupling plate **17h** via the rotary shaft **17m2** and is rotatable with respect to the coupling plate **17h**. The piston of the cylinder **17m1** is coupled to one end of the bar **17m4** via the rotary shaft **17m3** and is rotatable with respect to the bar **17m4**. The other end of the bar **17m4** is fixed with respect to one of a rotary shaft **17k3** or **17k4**. In the holding device **170**, the bar **17m4** is fixed with respect to the rotary shaft **17k3**. The second holder **2** is coupled with the other end of the link **17k1** and the other end of the link **17k2** respectively via rotary shafts **17k5** and **17k6** and is rotatable with respect to the links **17k1** and **17k2**.

(162) FIGS. **22A** and **22B** are side views schematically showing operations of the holding device according to the seventh modification of the embodiment. The cylinder **17m1**, the rotary shaft **17m2**, and the rotary shaft **17m3** are illustrated by double dot-dash lines in FIG. **22A**. In FIG. **22B**, the cylinder **17m1** and the rotary shaft **17m2** are not illustrated, and the rotary shaft **17m3** is illustrated by a dashed line. FIG. **22A** illustrates a state in which the second holder **2** is at the first position; and FIG. **22B** illustrates a state in which the second holder **2** is at the second position.

(163) The rotary shafts **17k3** to **17k6**, **17m2**, and **17m3** are parallel to the Y-direction. As illustrated in FIGS. **22A** and **22B**, the bar **17m4** rotates around the Y-direction with the rotary shaft **17k3** as a center when the cylinder **17m1** extends or contracts. Because the bar **17m4** is fixed with respect to the rotary shaft **17k3**, the rotary shaft **17k3** rotates according to the rotation of the bar **17m4**. The link **17k1** that is fixed to the rotary shaft **17k3** rotates around the Y-direction with the rotary shaft **17k3** as a center. The link **17k2** also rotates around the Y-direction synchronously with the link **17k1**. Thereby, the second holder **2** rotates around the Y-direction with the rotary shaft **17k3** as a center. The position in the X-Z plane of the second holder **2** can be changed by the operation of the drive mechanism **17m**. In the second state illustrated in FIG. **22B**, the gripping unit **2b** is positioned higher than the housing **1a** and the gripping unit **1b** of the first holder **1**.

(164) FIGS. **23A** to **23D** are side views schematically showing states when the suction hand according to the embodiment holds the article.

(165) FIGS. **23A** to **23C** illustrate the first operation. FIG. **23D** illustrates the second operation. When the article is gripped, the first holder **1** contacts the upper surface of the article **A** and grips the upper surface as shown in FIG. **23A**. Then, as shown in FIG. **23B**, the second holder **2** is caused to contact the side surface of the article **A** by the operation of the cylinder **17i**. The second holder **2** grips the side surface. The holding device **170** holds the article **A** by the gripping of the upper surface and the side surface of the article **A**.

(166) Continuing, the holding device **170** is raised. At this time, the cylinder **17c1** is extended by the weight of the first holder **1**, the second holder **2**, the article **A**, etc., when the cylinder **17c1** is released or the thrust due to the cylinder **17c1** is weak. Thereby, as shown in FIG. **23C**, the first

holder **1** and the second holder **2** are tilted with respect to the X-Y plane in the state in which the article A is held. There is a possibility that the contents of the article A may be broken if the tilt is too large. Therefore, it is favorable for the tilt angle with respect to the X-Y plane to be greater than 0 degrees and less than 15 degrees.

(167) For example, an adhesive for adhering a label, an adhesive of tape for packaging, etc., can be adhered to the upper surface of the article A. There are cases where the upper surface of another article A is stuck by the adhesive to the bottom surface of the article A that is held. By tilting, the article A that is held can be easily peeled from the other article A. Thereby, the friction between the other article A and the article A that is transferred can be reduced when transferring the article A.

(168) As shown in FIG. 23D, in the second operation, the holding device **170** also can grip the article A by attaching only the first holder **1** to the upper surface of the article A. At this time, the second holder **2** is positioned at the same level as the first holder or higher than at least a portion of the first holder **1**. Subsequently, similarly to FIG. 23C, the first holder **1** may be tilted with respect to the X-Y plane in the state in which the article A is gripped.

(169) According to the seventh modification, similarly to the holding device **100**, the first state in which the second holder **2** is at the first position and the second state in which the second holder **2** is at the second position can be switched. Therefore, the convenience of the holding device **170** can be improved. In the second state, similarly to the holding device **100**, the second holder **2** can be positioned higher than the first holder **1**. Thereby, the likelihood of interference with the other article A of the article A held by the second holder **2** can be further reduced.

(170) Eighth Modification

(171) FIG. 24 is a front view schematically illustrating a portion of a holding device according to an eighth modification of the embodiment.

(172) The holding device **180** according to the eighth modification differs from the holding device **170** in that multiple first holders **1** and multiple second holders **2** are included as shown in FIG. 24.

(173) Suction mechanisms **17z** that each include the first holder **1**, the second holder **2**, the linear guide **17d**, the coupling frame **17e**, the fixing mechanism **17f**, the linear guide **17g**, the coupling plate **17h**, the cylinder **17i**, the linkage **17k**, and the drive mechanism **17m** are located respectively at two side surfaces of the base **17b**. Thereby, the multiple first holders **1** and the multiple second holders **2** are arranged in the Y-direction.

(174) FIGS. 25 and 26 are front views schematically showing states when the holding device according to the eighth modification of the embodiment holds articles.

(175) The suction mechanisms **17z** are mutually-independently movable. The cylinders **17i** and **17m1** are independently operable for each suction mechanism **17z**. Therefore, as shown in FIGS. 25 and 26, the positions in the Z-direction of the first holders **1** may be different from each other. The positions in the X-Z plane of the second holders **2** may be different from each other.

(176) As shown in FIG. 25, multiple suction mechanisms **17z** may hold multiple articles A. As shown in FIG. 26, the multiple suction mechanisms **17z** may grip one article A. As an example, the shapes of the gripped articles A are rectangular parallelepipeds as shown in FIGS. 25 and 26. The upper surface of the article A may be tilted with respect to the X-Y plane. As shown in FIG. 26, the article of which the upper surface is tilted can be stably gripped by the positions in the Z-direction of the suction mechanisms **17z** changing according to the upper surface and by the suction pads **2d** deforming according to the position of the upper surface.

(177) According to the eighth modification, compared to the holding device **170**, diverse articles can be more stably held. The convenience of the holding device **180** can be further improved thereby.

(178) The forms described above can be combined as appropriate as long as the first state and the second state can be switched. For example, the fourth modification may be combined with the third modification. For example, both the first holder **1** and the second holder **2** may be movable in the vertical direction. The coupler **15** may be included in the holding devices **100**, **110**, **130**, **140**, **150**,

160, **170**, and **180** as in the holding device **120**. In the holding devices **100** to **140** and **160**, the tilts of the first and second holders **1** and **2** with respect to the X-direction may be changeable.

(179) An example is described above in which the first holder **1** and the second holder **2** hold the article by suction. The first holder **1** and the second holder **2** may hold the article by clamping. For example, the first holder **1** clamps two ends of the upper surface of the article. The second holder **2** clamps two ends of the side surface of the article. In such a case as well, the convenience of the holding device can be improved by the structures of the first and second holders **1** and **2** described above.

(180) FIG. **27** is a schematic view illustrating a hardware configuration.

(181) For example, the controller **50** has the hardware configuration illustrated in FIG. **27**. A processing device **90** illustrated in FIG. **27** includes a CPU **91**, ROM **92**, RAM **93**, a memory device **94**, an input interface **95**, an output interface **96**, and a communication interface **97**.

(182) The ROM **92** stores programs that control the operations of a computer. Programs that are necessary for causing the computer to realize the processing described above are stored in the ROM **92**. The RAM **93** functions as a memory region into which the programs stored in the ROM **92** are loaded.

(183) The CPU **91** includes a processing circuit. The CPU **91** uses the RAM **93** as work memory to execute the programs stored in at least one of the ROM **92** or the memory device **94**. When executing the programs, the CPU **91** executes various processing by controlling configurations via a system bus **98**.

(184) The memory device **94** stores data necessary for executing the programs and/or data obtained by executing the programs.

(185) The input interface (I/F) **95** connects the processing device **90** and an input device **95a**. The input I/F **95** is, for example, a serial bus interface such as USB, etc. The CPU **91** can read various data from the input device **95a** via the input I/F **95**.

(186) The output interface (I/F) **96** connects the processing device **90** and an output device **96a**. The output I/F **96** is, for example, an image output interface such as Digital Visual Interface (DVI), High-Definition Multimedia Interface (HDMI (registered trademark)), etc. The CPU **91** can transmit the data to the output device **96a** via the output I/F **96** and cause the output device **96a** to display an image.

(187) The communication interface (I/F) **97** connects the processing device **90** and a server **97a** that is outside the processing device **90**. The communication I/F **97** is, for example, a network card such as a LAN card, etc. The CPU **91** can read various data from the server **97a** via the communication I/F **97**. A camera **99** images articles and stores the images in the server **97a**. The camera **99** functions as the detector **35**.

(188) The memory device **94** includes not less than one selected from a hard disk drive (HDD) and a solid state drive (SSD). The input device **95a** includes not less than one selected from a mouse, a keyboard, a microphone (audio input), and a touchpad. The output device **96a** includes not less than one selected from a monitor and a projector. A device such as a touch panel that functions as both the input device **95a** and the output device **96a** may be used.

(189) According to embodiments described above, a convenient holding device, cargo handling apparatus, or holding method can be provided.

(190) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the invention. The above embodiments can be practiced in combination with each other.

Claims

1. A holding device, comprising: a first holder configured to hold an upper surface of an object; and a second holder configured to hold a side surface of the object, the holding device performing at least a first operation of the first and second holders holding the object, the first operation being performed in a first state in which the second holder can hold the side surface of the object, and a second operation of the first holder holding the object, the second operation being performed in a second state in which the second holder is positioned higher than at least a portion of the first holder, in a switching from the first state to the second state, as the second holder moves upward, a distance between the first holder and the second holder in a longitudinal direction, when viewed along a vertical direction, becomes shorter.
2. The device according to claim 1, further comprising: a first driver driving the first holder along the longitudinal direction; and a second driver driving the second holder along the longitudinal direction with respect to the first holder.
3. The device according to claim 1, wherein in the first state, the second holder is at a first position, and in the second state, the second holder is at a second position separated from the first position.
4. The device according to claim 1, wherein at least a portion of the second holder is positioned directly above the first holder in the second state.
5. The device according to claim 1, wherein the first state and the second state are switched by rotating the second holder around a rotation center, and the rotation center is along a lateral direction and is located at a different position from the second holder.
6. The device according to claim 1, wherein a plurality of the first holders and a plurality of the second holders are arranged in a lateral direction.
7. The device according to claim 6, wherein the plurality of first holders is mutually-independently movable in a vertical direction.
8. A cargo handling apparatus, comprising: a third driver driving the first and second holders in vertical direction; a lift on which the object is placed, the lift moving in the vertical direction; and the holding device according to claim 1.
9. The apparatus according to claim 8, further comprising: a first supporter and a second supporter supporting the first holder and the second holder, the first supporter and the second supporter being separated from each other in a lateral direction, a length in the lateral direction of the lift being greater than a length in the lateral direction between the first supporter and the second supporter.
10. The apparatus according to claim 9, wherein the first holder and the second holder are movable to a position not interfering with the lift when the lift is moved to be higher than the first holder.
11. The apparatus according to claim 8, further comprising: a controller, the controller receiving data of an arrangement of the object and switching a position of the second holder according to the data.
12. A holding device, comprising: a first holder configured to hold an upper surface of an object; and a second holder configured to hold a side surface of the object, the second holder being movable between a first position that is lower than at least a portion of the first holder, and a second position that is higher than the at least a portion of the first holder, at least a portion of the second holder being positioned directly above the first holder when the second holder is at the second position.
13. The device according to claim 12, further comprising: a fourth driver changing a position in a vertical direction of the second holder by moving the second holder with respect to the first holder.
14. A holding method, comprising: using a holding device to perform at least a first operation and a second operation, the holding device including a first holder configured to hold an upper surface of an article object, and a second holder configured to hold a side surface of the article object, only the first holder holding the object in the second operation, in the first operation, the first holder and the

second holder holding the object in a first state in which the second holder can hold the side surface of the object, in the second operation, the first holder holding the object in a second state in which the second holder is positioned higher than at least a portion of the first holder, in a switching from the first state to the second state, as the second holder moves upward, a distance between the first holder and the second holder in a longitudinal direction, when viewed along a vertical direction, becomes shorter.

15. The method according to claim 14, wherein: the second holder is positioned at a first position in the first operation, the second holder can hold the side surface of the object at the first position, and the second holder is positioned at a second position separated from the first position in the second operation.

16. The method according to claim 15, wherein the second position is higher than the first position.

17. The method according to claim 14, wherein the holding device is caused to receive data of an arrangement of the object, and switch a position of the second holder according to the data.
